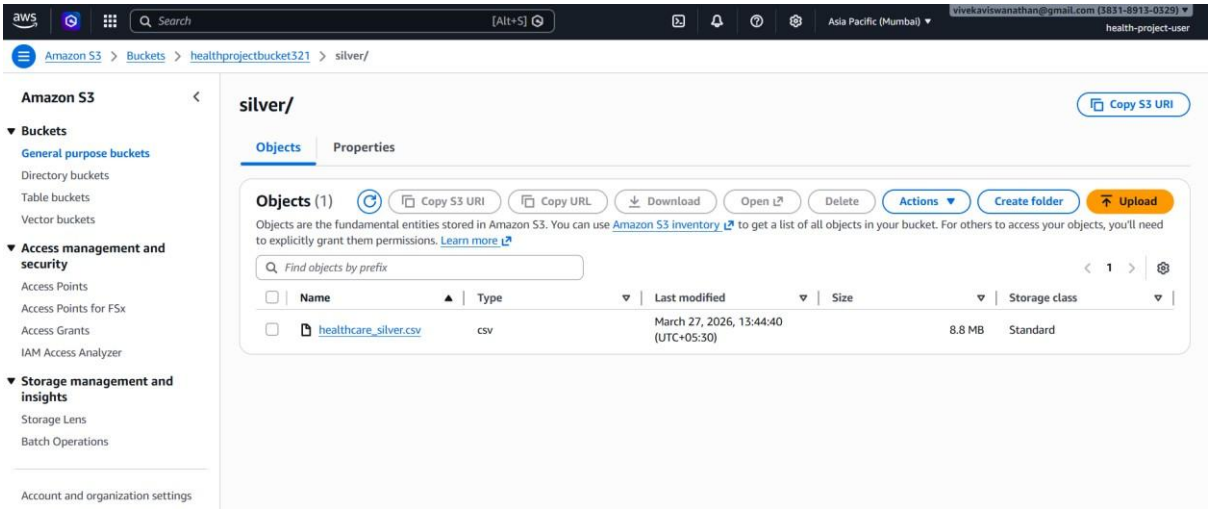
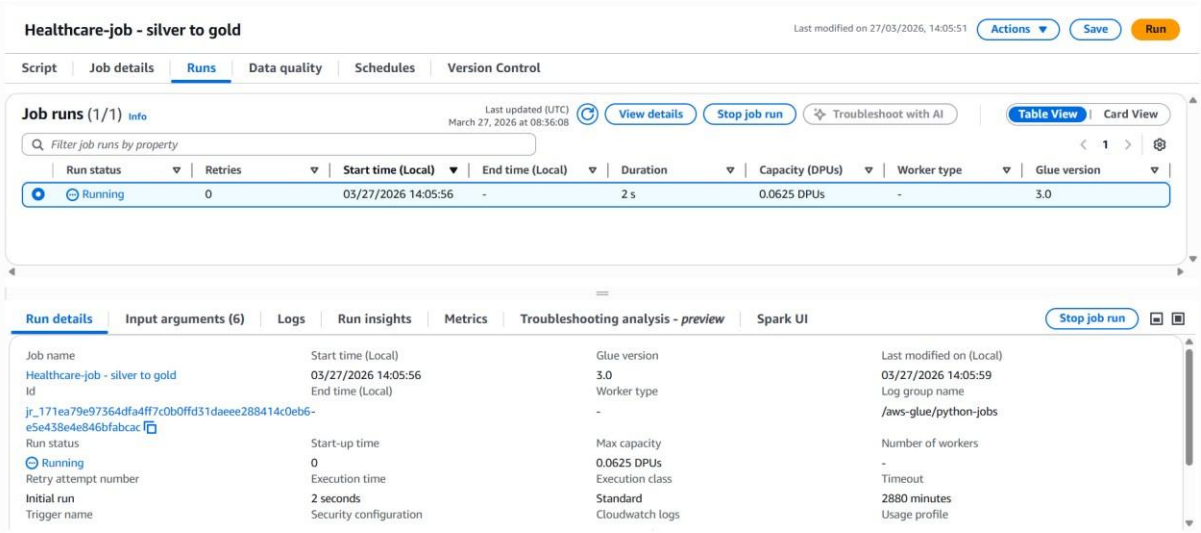
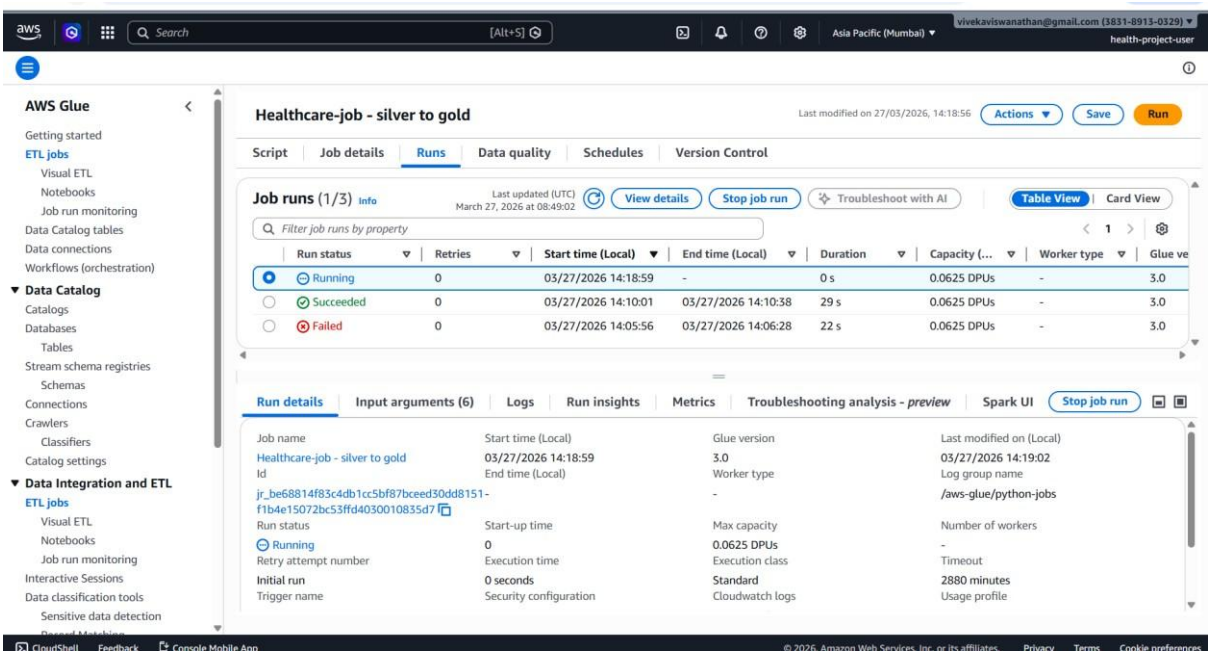
AWS – Healthcare Analytics



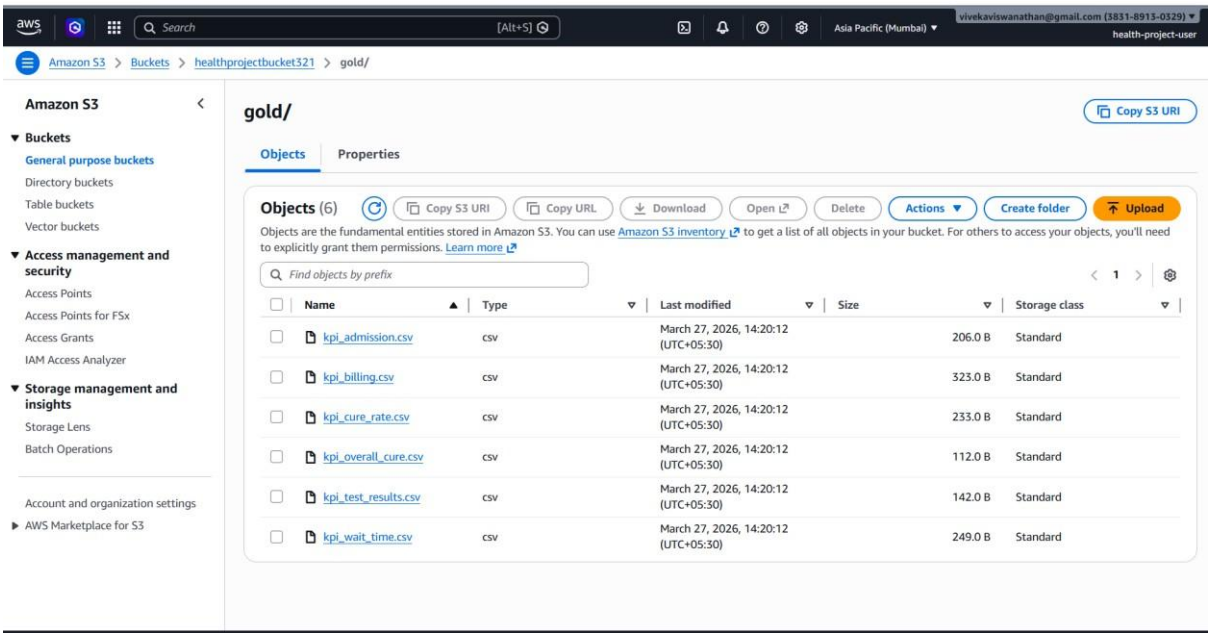
1)Uploaded healthcare_silver.csv to the S3 bucket healthprojectbucket321 under the silver/ layer.



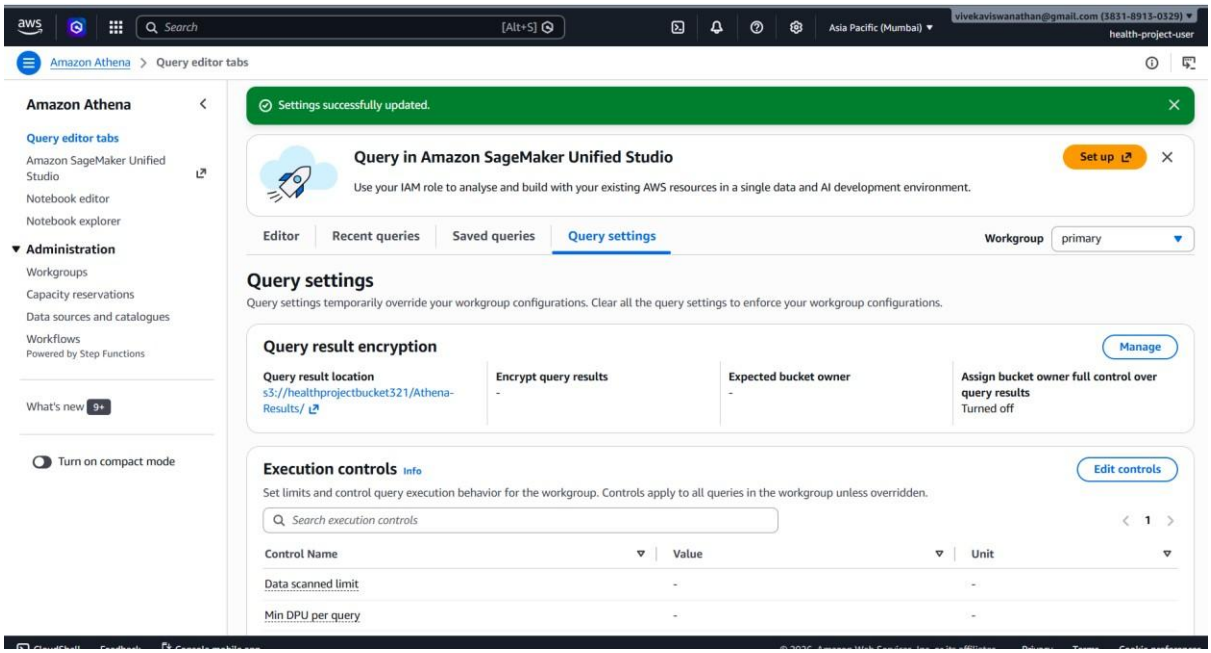
2)AWS Glue ETL job "Healthcare-job - silver to gold" was initiated and is currently running (first run, 0.0625 DPUs).

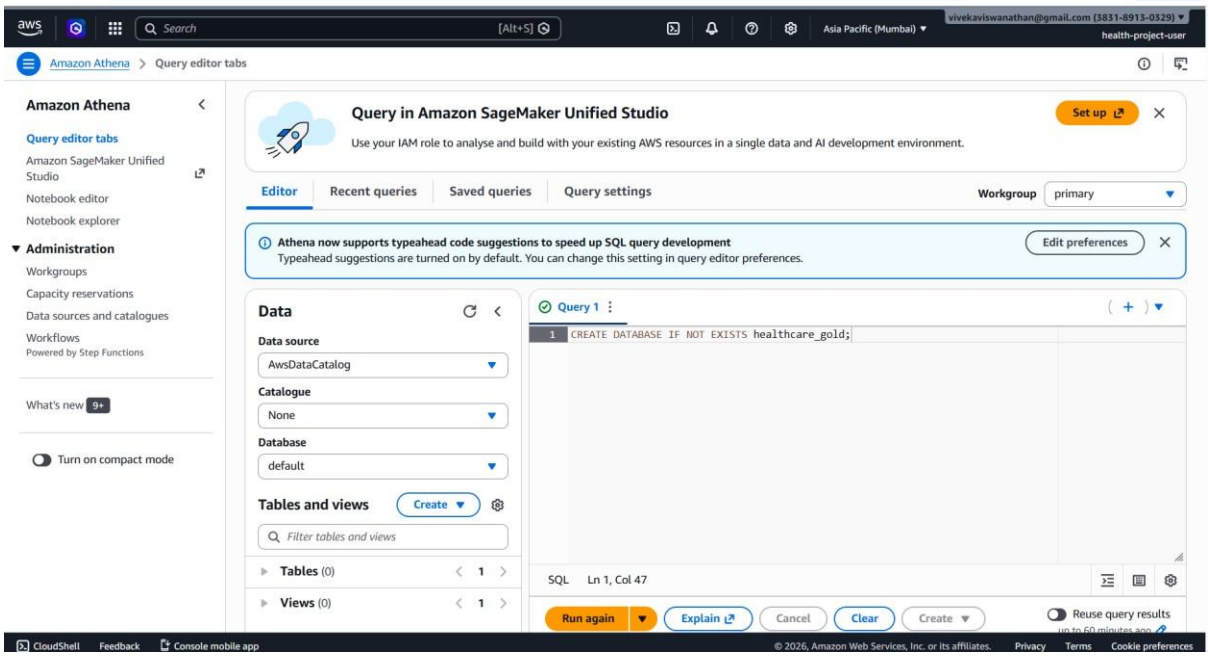


3)AWS Glue ETL job ran 3 times - one Running, one Succeeded (29s), and one Failed (22s) during the silver to gold transformation.



4)S3 gold/ folder confirmed with 6 KPI CSV files generated: kpi_admission, kpi_billing, kpi_cure_rate, kpi_overall_cure, kpi_test_results, and kpi_wait_time.





5)Amazon Athena Query Editor - executed "CREATE DATABASE IF NOT EXISTS healthcare_gold" to set up the gold database.

Amazon Athena Query settings configured with query result location set to s3://healthprojectbucket321/Athena-Results/.

Amazon Athena > Query editor tabs

Amazon Athena

- Query editor tabs
- Amazon SageMaker Unified Studio
- Notebook editor
- Notebook explorer

Administration

- Workgroups
- Capacity reservations
- Data sources and catalogues
- Workflows
  Powered by Step Functions

What's new 9+

Turn on compact mode

Athena now supports typeahead code suggestions to speed up SQL query development
Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.

Edit preferences

Data

Data source: AwsDataCatalog

Catalogue: None

Database: healthcare_gold

Tables and views Create

Filter tables and views

Tables (0) 1

Views (0) 1

Query 1 | Query 2

```
CREATE EXTERNAL TABLE IF NOT EXISTS
healthcare_gold.kpi_wait_time (
    Medical_Condition STRING,
    Avg_Length_of_Stay DOUBLE,
    Total_Patients INT
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
SKIP INPUT HEADER
LOCATION 's3://healthprojectbucket321/gold/kpi_wait_time.csv';
```

SQL Ln 1, Col 1

Run again | Explain | Cancel | Clear | Create | Reuse query results up to 60 minutes ago

Query results | Query status

line 1:8: mismatched input 'EXTERNAL'. Expecting: 'MATERIALIZED', 'MULTI', 'OR', 'PROTECTED', 'ROLE', 'SCHEMA', 'TABLE', 'VIEW'

Diagnose with Amazon Q

Amazon S3 > Buckets > healthprojectbucket321 > gold/

gold/ Copy S3 URI

Objects | Properties

Objects (6) Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class
kpi_admission/	Folder	-	-	-
kpi_billing/	Folder	-	-	-
kpi_cure_rate/	Folder	-	-	-
kpi_overall_cure/	Folder	-	-	-
kpi_test_results/	Folder	-	-	-
kpi_wait_time/	Folder	-	-	-

Amazon Athena > Query editor tabs

Settings successfully updated.

Editor | Recent queries | Saved queries | Query settings

Workgroup: primary

Query settings

Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.

Query result encryption Manage

Query result location	Encrypt query results	Expected bucket owner	Assign bucket owner full control over query results
s3://healthprojectbucket321/Athena-Results/	-	-	Turned off

Execution controls Info Edit controls

Set limits and control query execution behavior for the workgroup. Controls apply to all queries in the workgroup unless overridden.

Search execution controls

Control Name	Value	Unit
Data scanned limit	-	-
Min DPU per query	-	-
Max DPU per query	-	-

6)Amazon Athena Query settings successfully updated with the S3 result location and workgroup configurations.

S3 gold/ folder showing 6 KPI sub-folders created: kpi_admission/, kpi_billing/, kpi_cure_rate/, kpi_overall_cure/, kpi_test_results/, kpi_wait_time/.Amazon Athena - attempted CREATE EXTERNAL TABLE

kpi_wait_time but encountered a syntax error (mismatched input EXTERNAL).

Completed | Time in queue: 61 ms | Run time: 654 ms | Data scan

Results (6)

Copy | Downloa

Search rows

#	table_name	row_count
1	kpi_overall_cure	1
2	kpi_test_results	3
3	kpi_admission	3
4	kpi_wait_time	6
5	kpi_billing	5
6	kpi_cure_rate	6

Amazon Athena > Query editor tabs

Settings successfully updated.

Editor | Recent queries | Saved queries | Query settings | Workgroup primary

Query settings

Query settings temporarily override your workgroup configurations. Clear all the query settings to enforce your workgroup configurations.

Query result encryption

Manage

Query result location	Encrypt query results	Expected bucket owner	Assign bucket owner full control over query results
s3://healthprojectbucket321/Athena-Results/	-	-	Turned off

Execution controls Info

Edit controls

Set limits and control query execution behavior for the workgroup. Controls apply to all queries in the workgroup unless overridden.

Search execution controls

Control Name	Value	Unit
Data scanned limit	-	-
Min DPU per query	-	-
Max DPU per query	-	-

Amazon Athena > Query editor tabs

Data

Data source: AwsDataCatalog

Catalogue: None

Database: healthcare_gold

Tables and views | Create

Filter tables and views

Tables (6)

- kpi_admission
- kpi_billing
- kpi_cure_rate
- kpi_overall_cure
- kpi_test_results
- kpi_wait_time

Views (0)

Query 3 | Query 4 | Query 5 | Query 6 | Query 7 | Query 8 | Query 9

```sql
CREATE EXTERNAL TABLE IF NOT EXISTS
healthcare_gold.kpi_overall_cure (
    Total_Patients INT,
    Total_Cured INT,
    Total_Active INT,
    Overall_Cure_Rate DOUBLE,
    Avg_Wait_Days DOUBLE,
    Total_Monthly_Load_Mins DOUBLE
)
ROW FORMAT SERDE
'org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe'
WITH SERDEPROPERTIES ('field.delim' = ',')
STORED AS TEXTFILE
LOCATION 's3://healthprojectbucket321/gold/kpi_overall_cure/'
TBLPROPERTIES ('skip.header.line.count'='1');
```

SQL Ln 15, Col 46

Run again | Explain | Cancel | Clear | Create

Reuse query results up to 60 minutes ago

Query results | Query status

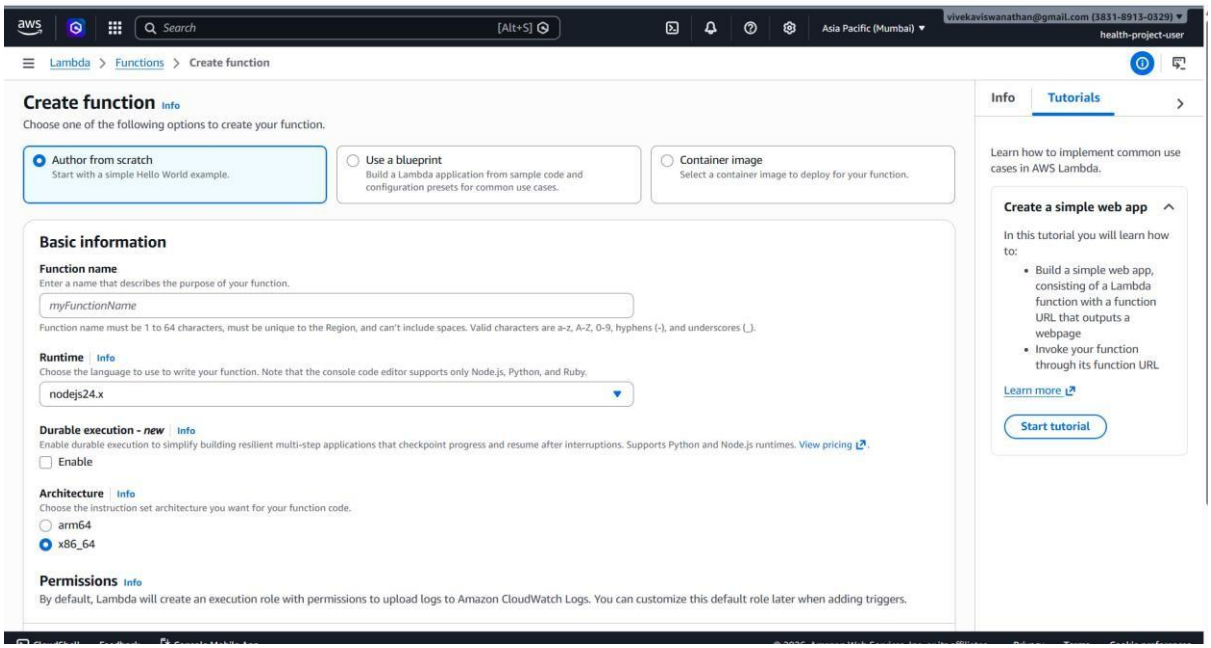
Completed | Time in queue: 41 ms | Run time: 324 ms | Data scanned: -

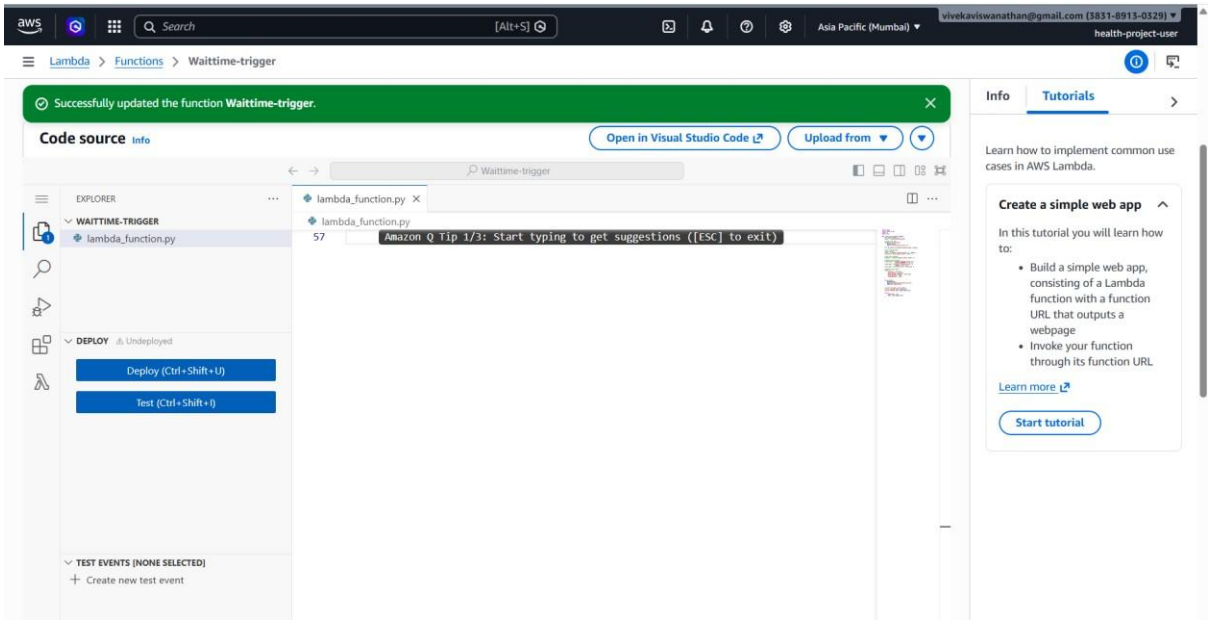
```
Query successful.
```

7)Amazon Athena - successfully created external table kpi_overall_cure using SERDE format; query completed successfully.

Amazon Athena Query settings page showing the configured primary workgroup and result location for query execution.

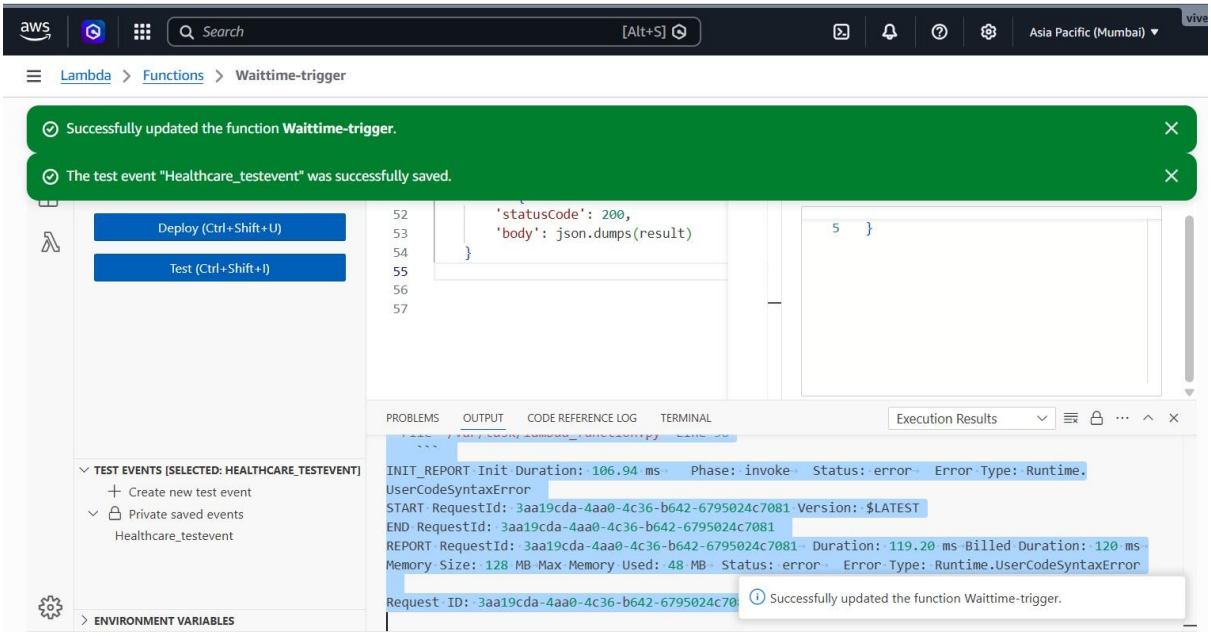
Athena query results confirmed 6 tables in the healthcare_gold database with their respective row counts.





8)AWS Lambda - Waittime-trigger function code (lambda_function.py) successfully deployed and updated.

AWS Lambda - Create Function page opened to author a new function from scratch using Python runtime.



*9)**AWS Lambda - Waittime-trigger encountered a UserCodeSyntaxError during the Healthcare_testevent test run.***

Error message

Status: Failed

Test Event Name: Healthcare_testevent

Response:

{

"errorMessage": "Unable to import module 'lambda_function': No module named 'pandas'",

"errorType": "Runtime.ImportModuleError",

"requestId": "",

"stackTrace": []

}

The area below shows the last 4 KB of the execution log.

Function Logs:

[WARNING] 2026-03-28T07:32:06.728Z LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda runtime update can cause unexpected function behavior. For functions using managed runtimes, runtime updates can be triggered by a function change, or can be applied automatically. To determine if the runtime has been updated, check the runtime version in the INIT_START log entry. If this error correlates with a change in the runtime version, you may be able to mitigate this error by temporarily rolling back to the previous runtime version. For more information, see https://docs.aws.amazon.com/lambda/latest/dg/runtimes-update.html

[ERROR] Runtime.ImportModuleError: Unable to import module 'lambda_function': No module named 'pandas'

Traceback (most recent call last):

INIT_REPORT Init Duration: 411.07 ms Phase: init Status: error Error Type: Runtime.ImportModuleError

[WARNING] 2026-03-28T07:32:07.019Z LAMBDA_WARNING: Unhandled exception. The most likely cause is an issue in the function code. However, in rare cases, a Lambda runtime update can cause unexpected function behavior. For functions using managed runtimes, runtime updates can be triggered by a function change, or can be applied automatically. To determine if the runtime has been updated, check the runtime version in the INIT_START log entry. If this error correlates with a change in the runtime version, you may be able to mitigate this error by temporarily rolling back to the previous runtime version. For more information, see https://docs.aws.amazon.com/lambda/latest/dg/runtimes-update.html

[ERROR] Runtime.ImportModuleError: Unable to import module 'lambda_function': No module named 'pandas'

Traceback (most recent call last):

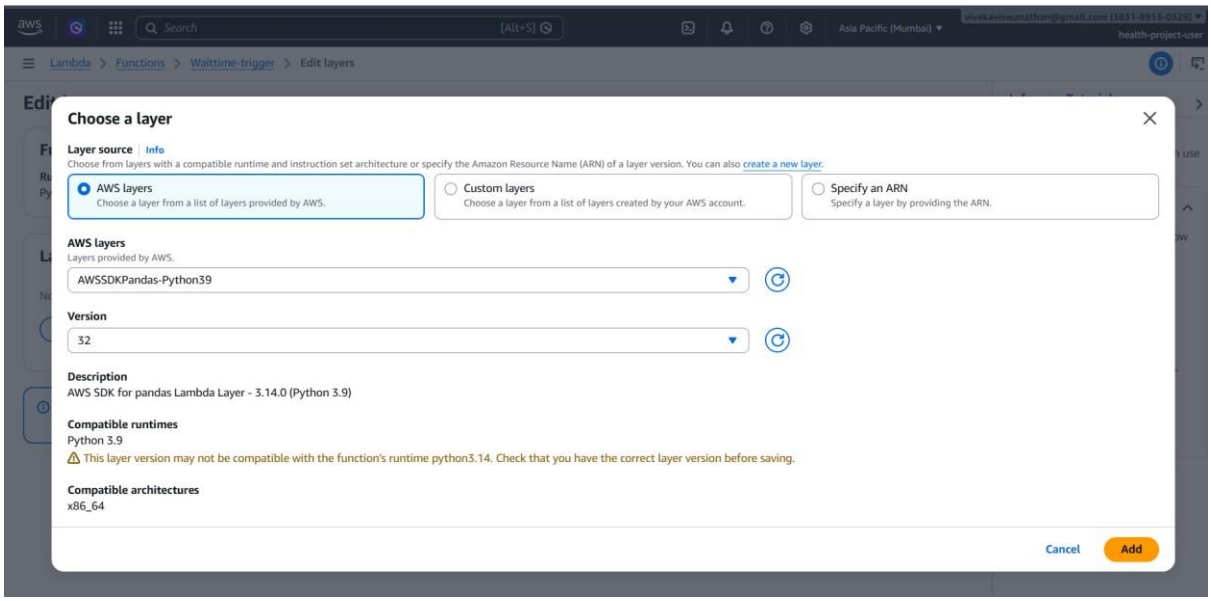
INIT_REPORT Init Duration: 275.48 ms Phase: invoke Status: error Error Type: Runtime.ImportModuleError

START RequestId: b4ce8fd6-1a0f-4adc-b463-932eadfc0ffd Version: $LATEST

END RequestId: b4ce8fd6-1a0f-4adc-b463-932eadfc0ffd

REPORT RequestId: b4ce8fd6-1a0f-4adc-b463-932eadfc0ffd Duration: 285.97 ms Billed Duration: 286 ms Memory Size: 128 MB Max Memory Used: 70 MB Status: error Error Type: Runtime.ImportModuleError

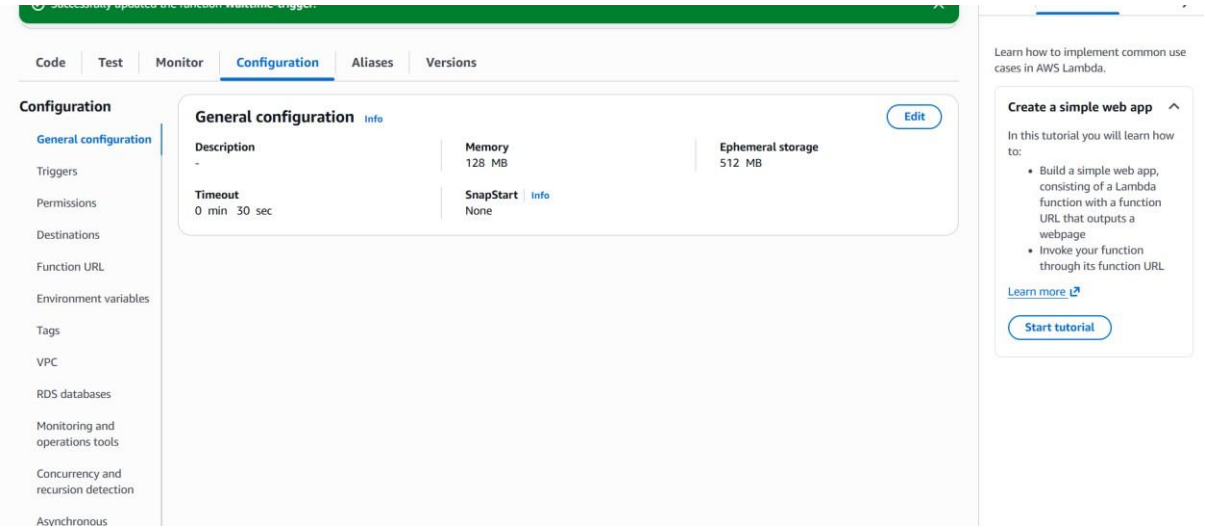
Request ID: b4ce8fd6-1a0f-4adc-b463-932eadfc0ffd




```
{
  "errorType": "Sandbox.Timedout",
  "errorMessage": "RequestId: 84ce5460-1608-43d4-8d12-96845dd0f211 Error: Task timed out after 3.00
  seconds"
}

The area below shows the last 4 KB of the execution log.

Function Logs:
START RequestId: 84ce5460-1608-43d4-8d12-96845dd0f211 Version: $LATEST
```

Ln 60, Col 1 Spaces: 4 UTF-8 CRLF Python Lambda Layout: US

```
The area below shows the last 4 KB of the execution log.

Function Logs:
START RequestId: 84ce5460-1608-43d4-8d12-96845dd0f211 Version: $LATEST
END RequestId: 84ce5460-1608-43d4-8d12-96845dd0f211
REPORT RequestId: 84ce5460-1608-43d4-8d12-96845dd0f211  Duration: 3000.00 ms    Billed Duration: 3000
ms    Memory Size: 128 MB Max Memory Used: 113 MB Status: timeout

Request ID: 84ce5460-1608-43d4-8d12-96845dd0f211
```



*10)**AWS Lambda Waittime-trigger General Configuration - Timeout set to 0 min 30 sec with 128 MB memory. Lambda execution log showing timeout status - Duration: 3000ms, Memory used: 113 MB, Status: timeout.***

Lambda execution log showing a Sandbox.Timeout error - the task timed out after 3.00 seconds.

AWS Lambda - Added AWSSDKPandas-Python39 layer (version 32) to the Waittime-trigger function to resolve the missing pandas module error.

PROBLEMS OUTPUT CODE REFERENCE LOG TERMINAL Execution Results

```
Test Event Name: Healthcare_testevent

Response:
{
  "statusCode": 200,
  "body": "{\"cure_rate\": 100.0, \"avg_wait_days\": 15.5, \"total_monthly_load_mins\": 181038.0,
  \"total_patients\": 54860, \"cured_patients\": 54860, \"active_patients\": 0}"
}

The area below shows the last 4 KB of the exec
```

No changes to deploy

Ln 60, Col 1 Spaces: 4 UTF-8 CRLF Python Lambda Layout: US

Trigger configuration Info

S3
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

s3/healthprojectbucket321

Bucket region: ap-south-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

All object create events

Prefix - *optional*
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.

e.g. images/

Suffix - *optional*
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.

e.g. .jpg

Recursive invocation
If your function writes objects to an S3 bucket, ensure that you are using different S3 buckets for input and output. Writing to the same bucket increases the risk of creating a recursive invocation, which can result in increased Lambda usage and increased costs. Learn more

I acknowledge that using the same S3 bucket for both input and output is not recommended and that this configuration can cause recursive invocations, increased Lambda usage, and increased costs.

Lambda will add the necessary permissions for AWS S3 to invoke your Lambda function from this trigger.

Info

Learn how to
cases in AWS

Create a

In this tut
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Learn mo

Start t

Lambda > Add triggers

S3
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

s3/healthprojectbucket321

Bucket region: ap-south-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with overlapping prefixes or suffixes that could match the same object key.

All object create events

Prefix - *optional*
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any special characters must be URL encoded.

silver/

Suffix - *optional*
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any special characters must be URL encoded.

.csv

Recursive invocation
If your function writes objects to an S3 bucket, ensure that you are using different S3 buckets for input and output. Writing to the same bucket increases the risk of creating a recursive invocation, which can result in increased Lambda usage and increased costs. Learn more

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Info

Info Tutorials

Learn how to impleme
cases in AWS Lambda.

Create a simple w

In this tutorial you w
to:

- Build a simple
consisting of a
function with
URL that outp
webpage
- Invoke your fu
through its fu

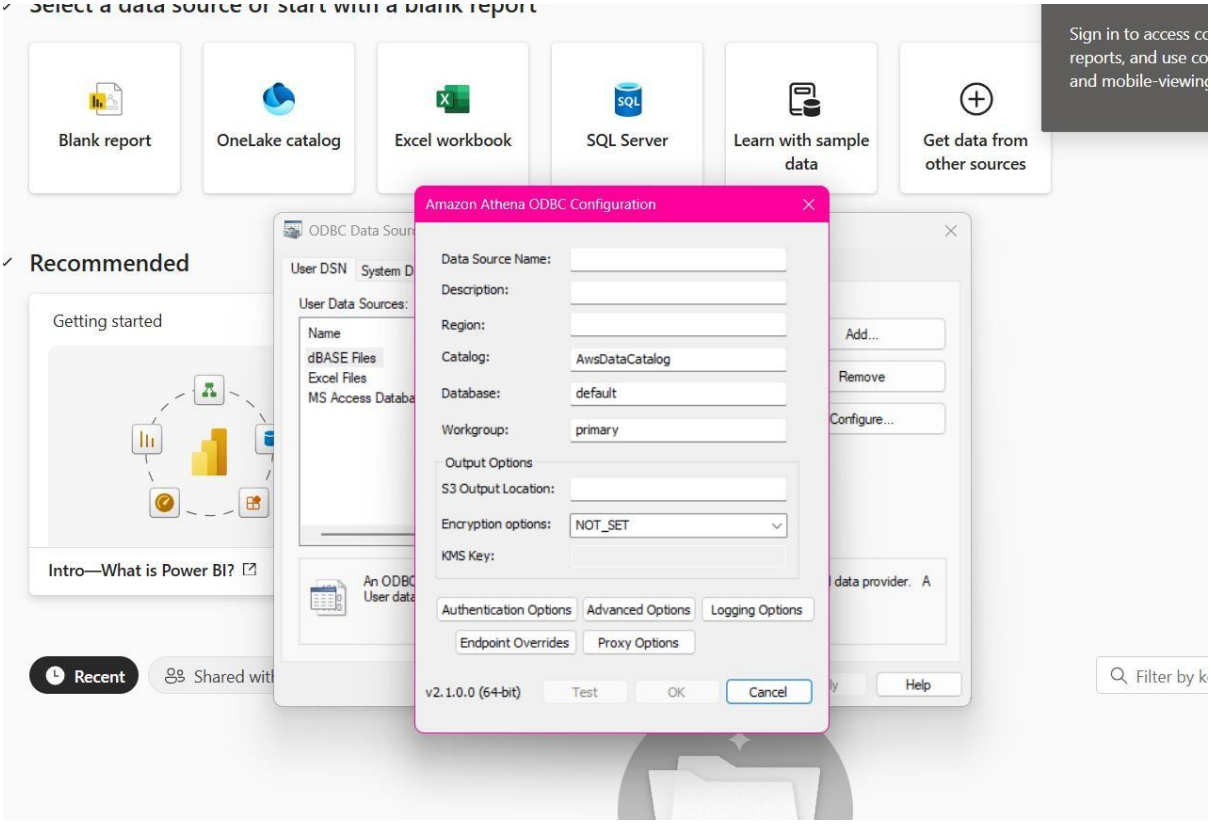
Learn more

Start tutorial

*11)**Lambda S3 trigger configured with prefix: silver/ and suffix: .csv to trigger on new CSV uploads to the silver layer.***

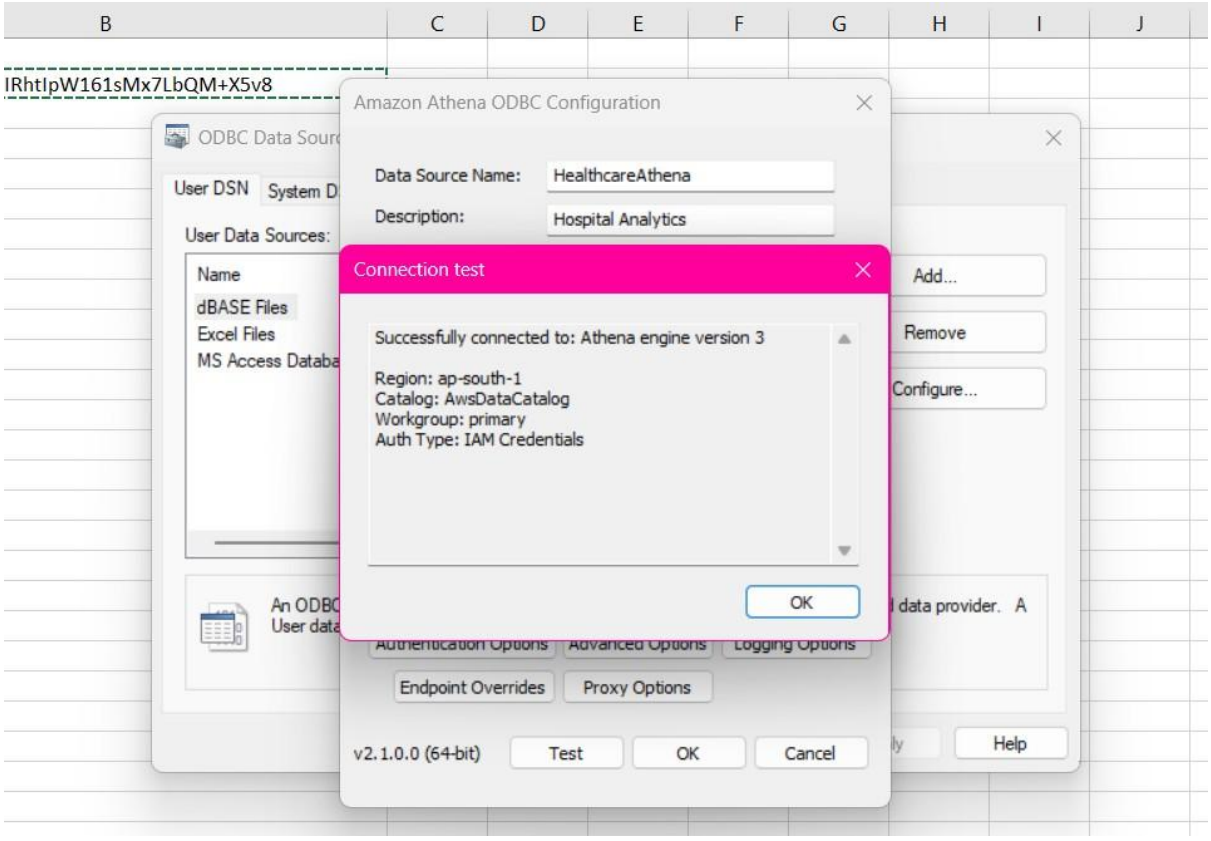
Lambda S3 Trigger configuration set up with bucket s3/healthprojectbucket321 for All object create events.

Lambda Healthcare_testevent returned a successful response (statusCode 200) with KPI data: cure_rate 100.0%, avg_wait_days 15.5, total_patients 54860.



12)Power BI - Amazon Athena ODBC Configuration set up with AwsDataCatalog catalog and primary workgroup.

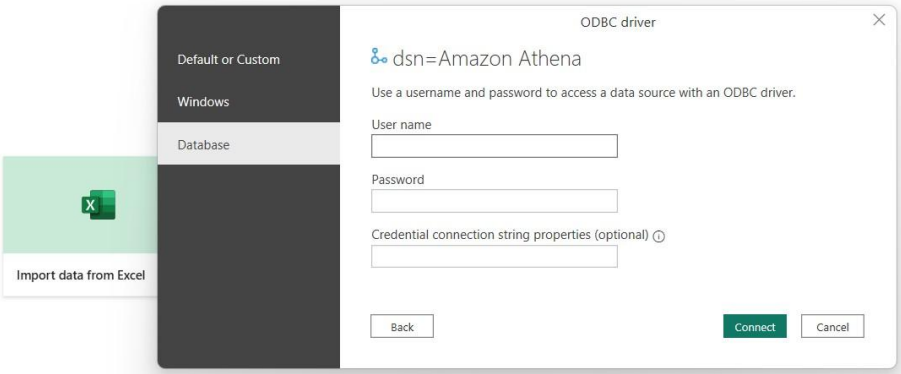
Connection failed: [AmazonAthena][AthenaClientError]: ExceptionName: UnrecognizedClientException, ErrorType: 17, ExceptionMessage: The security token included in the request is invalid.



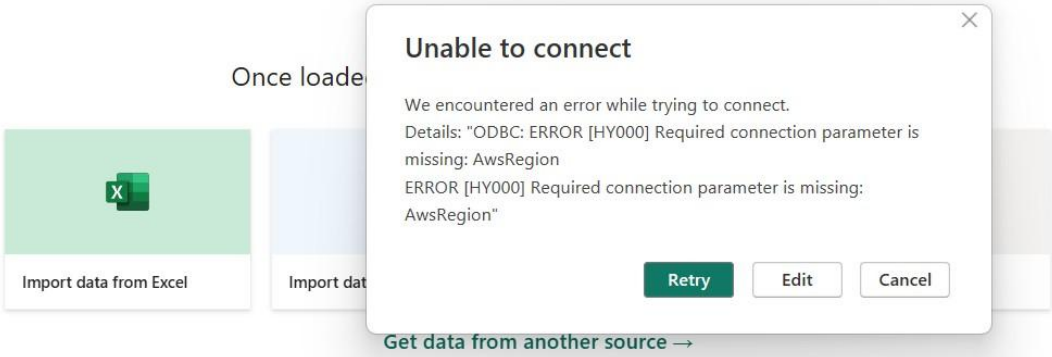
13)Amazon Athena ODBC Connection test successful - connected to Athena engine version 3, region ap-south-1 using IAM credentials.

Power BI

Error messages



Power BI ODBC driver - connecting via dsn=Amazon Athena with IAM credentials to import data.



14)Power BI - Unable to connect error: ODBC ERROR [HY000] Required connection parameter AwsRegion is missing from the configuration.

Uploaded healthcare_silver.csv to S3 silver layer, ran AWS Glue ETL job (Healthcare-job – silver to gold), and successfully generated 6 KPI CSV files (admission, billing, cure rate, overall cure, test results, wait time) in the S3 gold layer.